

02LQD Specification and Simulation of Digital Systems
2021/06/14

1. Combinational circuit design (2 points)

Design a combinational multiplier: input X is an integer in the range 0..5, output Z is $Z = 3X$. Define the number of input and output signals of the combinational circuit.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (16 points)

An FSM has a 1-bit input X and a 1-bit output Z . It serves as a sequence detector that searches for a series of binary inputs to satisfy the pattern $01[0^*]1$, where $[0^*]$ is a sequence of any number of consecutive zeroes, including none. The output (Z) should become true every time the sequence is found.

Example:

$X = 0\ 1\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 0\ 0\ 1$

$Z = 0\ 0\ 1\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 1$

1. Draw the state diagram of the Mealy Machine (**4 points**)
2. Draw the state diagram of the Moore Machine (**4 points**)
3. Design the Moore FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
4. Design the Mealy FSM resorting in Verilog with 2 always blocks (**4 points**).

3. HLSM (12 points)

A RAM has 1024 16-bit cells. A functional unit has a 1-bit input Go , a 1-bit output $Finish$, a 16-bit output Max , and the signals (to be defined by the candidate) required to access the RAM. The functional unit should read the RAM and output on Max the largest value. Operation starts when Go is asserted and $Finish$ is set to 1 when processing is over and the result is available on Max . Reset is synchronous. Design the functional unit as a HLSM, draw its state diagram (**6 points**) and write the VHDL code (**6 points**) to capture its behavior.

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

no later than Thursday June 17 11.59 pm

- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.